

VELS INSTITUTE OF SCIENCE, TECHNOLOGY AND ADVANCED STUDIES

DEPARTMENT: INFORMATION TECHNOLOGY

SEMESTER: I

TITLE OF THE COURSE:RDBMS

COURSE CODE:24CMIT13

TIME: 03 Hours

Max. Marks: 100

SECTION – A (3 MARKS)

S.No	Questions	Unit	K-Level	CO	PO
1.	What is a Database?	I	K1	CO1	PO1
2.	Define DBMS (Database Management System).	I	K1	CO1	PO1
3.	What is Data Redundancy in Database Systems?	I	K2	CO2	PO2
4.	What is the purpose of a Data Definition Language (DDL)?	I	K2	CO3	PO4
5.	List the types of data models used in DBMS.	I	K1	CO4	PO1
6.	What are the key differences between a DBMS and a File System?	I	K2	CO2	PO2
7.	What is meant by 'Data Independence' in DBMS?	I	K2	CO5	PO2
8.	List two advantages of using DBMS over traditional file systems.	I	K1	CO4	PO1
9.	What is the role of a Database Administrator (DBA)	I	K2	CO4	PO2
10.	What is the difference between data and information?	I	K1	CO2	PO1
11.	What are the different types of databases?	I	K1	CO2	PO1
12.	List any three applications of DBMS	I	K1	CO2	PO1
13.	What is a schema in DBMS?	I	K1	CO5	PO2
14.	What is an instance of a database?	I	K1	CO6	PO1
15.	What are the ACID properties of a transaction in a DBMS?	I	K1	CO5	PO1
16.	What are the three levels of the Three-Schema Architecture in DBMS?	I	K1	CO5	PO1
17.	Define the "External Schema" in the Three-Schema Architecture.	I	K2	CO5	PO2
18.	What is a Centralized DBMS architecture?	I	K1	CO6	PO1

19.	Define "Client" and "Server" in the context of a Client/Server DBMS architecture.	I	K2	CO6	PO2
20.	What are the advantages of using a Centralized DBMS?	I	K2	CO6	PO4
21.	What is an Entity in the context of an ER diagram?	II	K1	CO1	PO1
22.	Define Composite Attribute with an example.	II	K1	CO1	PO1,PO3
23.	What is a Primary Key ? Provide an example.	II	K1	CO3	PO1
24.	What is the difference between a Weak Entity and a Strong Entity ?	II	K2	CO1	PO1
25.	What does the Generalization process in ER modeling involve?	II	K2	CO1	PO2
26.	Define Composite Entity . Provide an example in the context of DBMS.	II	K1	CO3	PO3
27.	What is an Attribute in an ER diagram? Give an example.	II	K1	CO1	PO1
28.	Define Candidate Key with an example.	II	K1	CO3	PO1
29.	Explain the term Entity Set in the context of ER modeling.	II	K1	CO1	PO1,PO2
30.	What is the difference between Super Key and Primary Key ?	II	K2	CO3	PO1
31.	What is the difference between Generalization and Specialization in ER modeling?	II	K2	CO1	PO2
32.	What is a Weak Entity Set ? Give an example.	II	K1	CO1	PO1
33.	What is a Multivalued Attribute ? Provide an example.	II	K1	CO1	PO1,PO3
34.	Define Relational Schema . What is its role in DBMS?	II	K1	CO3	PO1
35.	What is Cardinality in an ER diagram?	II	K1	CO1	PO1
36.	What is a tuple?	II	K1	CO1	PO1
37.	What is a relation?	II	K1	CO2	PO1
38.	Define Foreign Key Constraint in a relational database.	II	K1	CO2	PO1PO,3
39.	What is a super key?	II	K1	CO2	PO1
40.	What is a Transaction in DBMS? Explain briefly.	II	K1	CO2	PO4
41.	What does the term "Normalization" mean in relational databases?	III	K2	CO2	PO1
42.	What does 1NF (First Normal Form) mean in relational databases?	III	K2	CO1	PO1
43.	What is the purpose of using a Candidate Key in a relational schema?	III	K2	CO2	PO1

44.	Explain what is meant by 'Transitive Dependency' in relational databases.	III	K2	CO2	PO1
45.	What is Boyce-Codd Normal Form (BCNF)?	III	K2	CO1	PO1
46.	What is the purpose of normalization in database design?	III	K1	CO1	PO1
47.	Define the concept of Functional Dependency in the context of relational databases.	III	K2	CO3	PO2
48.	What is 1NF and how does it resolve redundancy in a database?	III	K1	CO2	PO1
49.	How does 2NF differ from 1NF in terms of database normalization?	III	K2	CO2	PO2
50.	Explain why a relation in 3NF does not have any Transitive Dependencies .	III	K2	CO3	PO4
51.	What is meant by Redundancy in a database? How does normalization help in reducing it?	III	K1	CO1	PO1
52.	How do you determine whether a relation is in 1NF ? Provide a simple example.	III	K1	CO2	PO2
53.	Define Partial Dependency and explain why it violates 2NF .	III	K1	CO3	PO2
54.	What is the difference between Partial Dependency and Transitive Dependency ?	III	K2	CO3	PO1
55.	Explain the term Atomic Attribute in the context of 1NF .	III	K1	CO2	PO1
56.	What does BCNF ensure that 3NF might not? Provide an example where a relation is in 3NF but not in BCNF .	III	K1	CO4	PO1
57.	What is the significance of a Primary Key in database normalization?	III	K1	CO2	PO1
58.	How does normalization in database design help in eliminating Insertion Anomalies ?	III	K2	CO1	PO4
59.	What is Multi-valued Dependency (MVD) , and how does it affect a relation ?	III	K1	CO4	PO2
60.	What is an anomaly in RDBMS?	III	K1	CO2	PO1
61.	Define SQL?	IV	K1	CO1	PO1
62.	What are the types of SQL Commands?	IV	K1	CO1	PO1
63.	Define DDL Commands?	IV	K2	CO2	PO2
64.	What is Data Types and its Types?	IV	K2	CO3	PO4
65.	What is Primary Key?	IV	K1	CO4	PO1
66.	Define SELECT Statement?	IV	K2	CO2	PO2
67.	What is Modification?	IV	K2	CO5	PO2

68.	Define Truncation?	IV	K1	CO4	PO1
69.	Define Constraints?	IV	K2	CO4	PO2
70.	Explain about Subquery?	IV	K1	CO2	PO1
71.	What is ORDERBY and GROUPLY?	IV	K1	CO2	PO1
72.	Define Join and Set Operation?	IV	K1	CO2	PO1
73.	What is MySQL?	IV	K1	CO5	PO2
74.	What is Integrity Constraints?	IV	K1	CO6	PO1
75.	Define TCL Commands?	IV	K1	CO5	PO1
76.	What is Projection?	IV	K1	CO5	PO1
77.	What is DCL Commands?	IV	K2	CO5	PO2
78.	What is Aggregate Functions?	IV	K1	CO6	PO1
79.	What is Foreign Key?	IV	K2	CO6	PO2
80.	Define not Null?	IV	K2	CO6	PO4
81.	What is Function?	V	K1	CO1	PO1
82.	Define Aggregate Function?	V	K1	CO1	PO1,PO3
83.	What is String Function?	V	K1	CO3	PO1
84.	Define Math Function?	V	K2	CO1	PO1
85.	Define Data?	V	K2	CO1	PO2
86.	Define Time Function?	V	K1	CO3	PO3
87.	What is Joins?	V	K1	CO1	PO1
88.	What is Sub queries in MySQL?	V	K1	CO3	PO1
89.	What is MySQL Control statement?	V	K1	CO1	PO1,PO2
90.	What is MySQL Triggers?	V	K2	CO3	PO1
91.	What is Stored procedures?	V	K2	CO1	PO2
92.	Define Inner Join?	V	K1	CO1	PO1
93.	Define Right Join?	V	K1	1	PO1,PO3

94.	Define Left Join?	V	K1	CO3	PO1
95.	Define Cross Join?	V	K1	CO1	PO1
96.	Define Full Join?	V	K1	CO1	PO1
97.	Write a syntax for Right Join?	V	K1	CO2	PO1
98.	Write a syntax for Aggregate Functions?	V	K1	CO2	PO1,PO3
99.	Write a example for Alter?	V	K1	CO2	PO1
100.	Write a syntax for Update?	V	K1	CO2	PO4

SECTION – B (8 MARKS)

S.No	Questions	Unit	K Level	CO	PO
101.	Explain the architecture of a Database Management System (DBMS).	I	K3	CO4	PO1
102.	Describe the different types of data models in DBMS. Explain their advantages and disadvantages.	I	K3	CO4	PO2
103.	Explain the concept of Data Independence and differentiate between Logical and Physical Data Independence .	I	K2	CO1	PO1
104.	What are the key advantages of using DBMS over traditional file systems? Discuss with examples.	I	K3	CO2	PO2
105.	Describe the three-schema architecture of a DBMS and explain the role of each schema.	I	K2	CO1	PO1
106.	List and explain the types of database models with examples.	I	K2	CO3	PO2
107.	What is the Client-Server Architecture for DBMS? Explain its components and working.	I	K3	CO2	PO1,PO5
108.	Define DBMS and explain its role in managing data with respect to data redundancy, integrity, and security .	I	K2	CO2	PO4
109.	Explain the Relational Model of data with its key components such as tables, keys, and constraints.	I	K2	CO3	PO1
110.	Describe the ACID properties in DBMS and explain their importance in database transactions.	I	K2	CO2	PO1,PO2
111.	Define a DBMS schema and explain the differences between internal, conceptual, and external schemas in detail.	I	K2	CO1	PO1
112.	Explain Distributed Database Systems . What are the challenges and benefits of using distributed DBMS over centralized DBMS?	I	K3	CO1	PO1

113.	Explain the concept of Backup and Recovery in DBMS. Why is it important, and what are the different strategies?	I	K2	CO2	PO4,PO5
114.	Explain the concept of Concurrency Control in DBMS. Discuss the different techniques used for managing concurrent transactions.	I	K3	CO2	PO1,PO4
115.	Explain the difference between DDL, DML, and DCL with examples of each.	I	K2	CO2	PO1
116.	What is Indexing in DBMS? Discuss the different types of indexes (e.g., single-level, multi-level) and their advantages.	I	K2	CO2	PO1
117.	What is a Super Key? Explain its significance in DBMS with an example.	II	K2	CO2	PO1
118.	Differentiate between Primary Key and Candidate Key.	II	K3	CO3	PO1
119.	Explain the concept of Strong Entity & Weak Entity with an example.	II	K2	CO2	PO2
120.	What is the difference between Generalization and Specialization in ER modeling?	II	K3	CO3	PO1
121.	Explain the term Composite Key with an example.	II	K2	CO2	PO1
122.	What is an Alternate Key? How does it differ from the Primary Key?	II	K2	CO2	PO1
123.	How is a Weak Entity identified in an ER diagram? Provide an example.	II	K2	CO2	PO2
124.	Describe the steps involved in mapping an ER model to a relational schema.	II	K3	CO3	PO1
125.	What is an Entity in an ER diagram? Provide an example of an entity and its attributes.	II	K2	CO2	PO1
126.	Define a Composite Attribute with an example. How is it represented in an ER diagram?	II	K2	CO2	PO1
127.	What are Multi-valued Attributes in an ER model? Provide an example.	II	K2	CO2	PO1
128.	Explain the concept of a Foreign Key and its role in relational databases.	II	K2	CO2	PO1
129.	What is the difference between a Simple Attribute and a Composite Attribute?	II	K2	CO2	PO1
130.	Explain how a One-to-Many relationship is represented in an ER diagram and later mapped to a relational schema.	II	K3	CO3	PO1
131.	Describe the concept of a Primary Key in relational databases. How is it different from a Candidate Key?	II	K2	CO2	PO1
132.	What is a relationship set in an ER diagram? Give an example of a relationship set.	II	K2	CO2	PO1
133.	What is an example of a Multi-valued Dependency (MVD)?	III	K2	CO2	PO1

134.	What is the role of a Composite Key in a relational schema? Provide an example where a composite key is necessary.	III	K2	CO2	PO1
135.	Explain the concept of Aggregation in an ER diagram. How does it differ from Generalization and Specialization?	III	K2	CO2	PO1
136.	What is the process of decomposing a relation to achieve Boyce-Codd Normal Form (BCNF) ? Provide an example of a table that violates BCNF and how to decompose it.	III	K3	CO3	PO1
137.	Explain Partial Dependency in the context of relational databases. How does it relate to Second Normal Form (2NF) ?	III	K2	CO2	PO1
138.	What is a Multi-valued Attribute ? Explain with an example and how to map it to a relational schema.	III	K2	CO2	PO1
139.	Define a Transitive Dependency and explain how it affects Third Normal Form (3NF) . Provide an example of a table violating 3NF and how to resolve it.	III	K3	CO3	PO1
140.	What are Functional Dependencies and why are they important in database design? Provide an example of a functional	III	K3	CO3	PO2
141.	Discuss the significance of Normalization in relational database design. Why is 2NF important for reducing redundancy and ensuring data integrity?	III	K2	CO2	PO1
142.	What is 3NF (Third Normal Form) ? Explain how 3NF eliminates redundancy and ensures data consistency. Provide an example of a table that is not in 3NF and how to convert it to 3NF.	III	K3	CO3	PO1
143.	Describe the process of mapping a weak entity to a relational schema. Provide an example where a weak entity needs to be mapped.	III	K3	CO3	PO1
144.	What is an anomaly in RDBMS ?Explain it with example.	III	K1	CO2	PO2
145.	What is the purpose of Normalization ? Explain how 2NF (Second Normal Form) improves upon 1NF (First Normal Form) .	III	K2	CO2	PO1
146.	Discuss the concept of Transitive Dependency and explain how it violates 3NF . Provide an example and how to eliminate this dependency.	III	K3	CO3	PO1
147.	What is the significance of Redundancy in a database? How does normalization address the issue of redundancy, and why is it important?	III	K2	CO2	PO1
148.	What is the significance of Atomicity in the context of First Normal Form (1NF)? How does it impact database design?	III	K2	CO2	PO1

149.	What is the concept of Functional Dependency in a relational schema? How does it affect database normalization?	IV	K2	CO2	PO1
150.	What is SQL and its statement?	IV	K3	CO4	PO1
151.	What are all the most important commands in SQL?	IV	K3	CO4	PO2
152.	Explain about SQL SELECT statement?	IV	K2	CO1	PO1
153.	What is Numeric Data Types in MySQL?	IV	K3	CO2	PO2
154.	Explain about String data type in MYSQL?	IV	K2	CO1	PO1
155.	Explain DDL Commands vs DML Commands?	IV	K2	CO3	PO2
156.	Explain about Joins?	IV	K3	CO2	PO1,PO5
157.	Explain about features s of MYSQL?	IV	K2	CO2	PO4
158.	Difference between SQL and MYSQL?	IV	K2	CO3	PO1
159.	Explain about Aggregate Function?	IV	K2	CO2	PO1,PO2
160.	Explain and types of integrity constraints?	IV	K2	CO1	PO1
161.	Explain about DCL Commands in SQL?	IV	K3	CO1	PO1
162.	What are the SELECT statement with CLAUSES?	IV	K2	CO2	PO4,PO5
163.	Explain about DML?	IV	K3	CO2	PO1,PO4
164.	Explain about Selection , Projection?	IV	K2	CO2	PO1
165.	Explain about COUNT Function?	V	K2	CO2	PO1
166.	Explain about SUM Function?	V	K2	CO2	PO1
167.	Explain about AVG and MAX function?	V	K3	CO3	PO1
168.	Explain about MIN Function?	V	K2	CO2	PO2
169.	Explain about Aggregate Function?	V	K3	CO3	PO1
170.	Explain anout MySQL control statement?	V	K2	CO2	PO1
171.	Explain about String Function?	V	K2	CO2	PO1
172.	Explain about Stored procedure?	V	K2	CO1	PO1
173.	Explain with syntax for date and time function?	V	K3	CO1	PO1

174.	Explain about Math function with syntax and example?	V	K3	CO2	PO2
175.	Write syntax and example for control statement?	V	K2	CO1	PO1
176.	Explain about MySQL Triggers?	V	K2	CO3	PO2
177.	Write a program for Aggregate Function?	V	K3	CO2	PO1,PO5
178.	Write a program for Control statement?	V	K2	CO2	PO4
179.	Write a program for MySQL Triggers?	V	K2	CO3	PO1
180.	Write a program for Math functions?	V	K2	CO2	PO1,PO2

SECTION – C (15 MARKS)

S.No	Questions	Unit	K Level	CO	PO
181.	Discuss the Relational Model of Data in detail. Explain its components (tables, tuples, attributes, primary keys, foreign keys, etc.), with examples.	I	K3	CO3	PO1,PO3
182.	Describe the Entity-Relationship (ER) Model and illustrate how to convert an ER diagram to a relational schema with a real-world example.	I	K3	CO1	PO2,PO3
183.	What are the ACID properties of database transactions? Explain each property with examples and describe why they are important for ensuring the reliability of a DBMS.	I	K3	CO2	PO1,PO4
184.	Discuss the concept of indexing in DBMS. Explain different types of indexes (single-level, multi-level, clustered, and non-clustered) with their advantages and disadvantages.	I	K3	CO2	PO5,PO3
185.	Explain the concept of Concurrency Control in DBMS. Discuss the different techniques used for managing concurrent transactions	I	K3	CO2	PO1,PO4
186.	Describe the Client-Server Architecture in DBMS. Explain its components and the communication mechanism between the client and the server. Discuss the advantages and challenges of this architecture.	I	K3	CO2	PO1,PO5
187.	Discuss the Backup and Recovery process in DBMS. Explain various types of backups (full, incremental, differential) and recovery techniques (rollback, rollforward).	I	K3	CO2	PO4
188.	Define Distributed Database Systems . Discuss the types of distributed databases (homogeneous, heterogeneous) and explain the challenges in managing distributed databases.	I	K3	CO2	PO1
189.	Discuss the types of relationships in ER diagrams with examples.	II	K3	CO1	PO1

190.	Explain the concept of mapping EER model constraints (specialization) to relational schemas.	II	K3	CO3	PO1
191.	What is a Relational Schema? Explain how Entity-Relationship models are mapped to Relational Schema.	II	K3	CO3	PO1
192.	Describe the process of converting an ER diagram to a relational database design, including how relationships and attributes are mapped.	II	K3	CO3	PO1
193.	Explain the different types of keys (Super Key, Primary Key, Candidate Key, Alternate Key) with examples.	II	K2	CO2	PO1
194.	What are the differences between a One-to-Many and a Many-to-Many relationship? Provide examples and explain how each is mapped to a relational schema.	II	K3	CO3	PO1
195.	Discuss the process of converting an Enhanced ER (EER) diagram into a relational schema. Include mapping of generalization, specialization, and inheritance.	II	K3	CO3	PO1
196.	Explain the various types of attributes in an ER model with examples. How are they mapped to relational schema?	II	K3	CO3	PO1
197.	Describe the concept of Normalization . Explain the various normal forms (1NF, 2NF, 3NF, BCNF) with examples, and describe the steps involved in converting a relation into each normal form.	III	K3	CO3	PO1
198.	Discuss the role and significance of Functional Dependencies in relational database design. Provide examples of each type of functional dependency (full, partial, transitive) and explain their impact on normalization.	III	K2	CO2	PO1
199.	What are Superkeys , Candidate Keys , and Primary Keys in a relational database? Explain their definitions, differences, and their roles in ensuring data integrity. Provide examples for each.	III	K2	CO2	PO1
200.	Explain the process of decomposing a relation into smaller relations to achieve Boyce-Codd Normal Form (BCNF) . Illustrate the steps involved with an example of a relation that violates BCNF and how it can be decomposed into BCNF.	III	K3	CO3	PO1
201.	What is the significance of Atomicity in the context of First Normal Form (1NF) ? Explain how the violation of atomicity leads to issues in a relational database design. Provide an example where atomicity is violated and how it can be resolved.	III	K2	CO2	PO1

202.	Explain the process of mapping multivalued attributes in an ER diagram to a relational schema. What challenges are faced during the mapping, and how can these challenges be overcome? Provide a specific example.	III	K3	CO3	PO1
203.	Explain Hierarchical Data Models and compare them with Relational Data Models . Discuss the advantages and disadvantages of using a hierarchical model over a relational model for database design.	III	K2	CO2	PO1
204.	Define Transaction Management in relational databases. Explain the ACID properties (Atomicity, Consistency, Isolation, Durability) and their role in ensuring reliable transactions.	III	K2	CO2	PO1
205.	Write a program for Table Creation, Renaming a Table, Copying another table, Dropping a Table.	IV	K2	CO2	PO1
206.	Explain about popular DDL Commands with example for CREATE,ALTER,DROP,TRUNCATE,COMMENT	IV	K3	CO3	PO1
207.	Explain about SET Operations in SQL?	IV	K2	CO2	PO1
208.	Explain five key features and benefits of MYSQL?	IV	K2	CO2	PO1
209.	Briefly explain about Integrity Constraints?	IV	K3	CO3	PO1
210.	Explain about Join Operation?	IV	K3	CO3	PO2
211.	Explain about DML,DCL,TCL Commands?	IV	K2	CO2	PO1
212.	Explain about Features of MySQL?	IV	K3	CO3	PO1
213.	Explain about Function in MySQL?	V	K2	CO2	PO1
214.	Briefly explain about Aggregate functions and String Functions?	V	K2	CO2	PO1
215.	Explain about Date and Time Functions?	V	K3	CO3	PO1
216.	Explain about MySQL Triggers?	V	K2	CO2	PO1
217.	Explain about MySQL Control statements?	V	K3	CO3	PO1
218.	What is ORDERBY and GROUPBY?	V	K2	CO2	PO1
219.	Explain about Math Functions?	V	K2	CO2	PO1
220.	Write about string function?	V	K2	CO2	PO1

K1- Remember; K2- Understand; K3- Apply; K4- Analyze;

K5-Evaluate; K6-Create